GROUP G12 · BASELINE REPRODUCTION REPORT

Reproducing Sustainable LLM Inference for Edge AI on a Consumer GPU

A reproduction of Husom et al. (ACM TIOT, 2025), the research gap it exposes, and a thermal-aware estimator that cuts sustained-throughput prediction error from 51% to 11%.

GROUP	PRIMARY RESEARCHER	CO-RESEARCHER	RESEARCH AREA
G12 · Section A	Faheem · 23i-0728	Irtaza Kazmi · 23i-6001	Edge–Cloud Intelligence

01 Selected paper

Title	Sustainable LLM Inference for Edge AI: Evaluating Quantized LLMs for Energy Efficiency, Output Accuracy, and Inference Latency
Authors	E. J. Husom, A. Goknil, M. Astekin, L. K. Shar, A. Kåsen, S. Sen, B. A. Mithassel, A. Soylu
Venue	ACM Transactions on Internet of Things, 6(4) — peer-reviewed journal
Year	2025 (preprint April 2025, published November 2025)
DOI	10.1145/3767742 · preprint arXiv:2504.03360
Code	github.com/ejhusom/MELODI
Data & scripts	github.com/ejhusom/neurips-edge-llm-challenge-sampled
Commits used	MELODI `d55dae9` · data `c11fe13`

PROBLEM ADDRESSED

Running LLMs on edge devices is attractive for cost, latency and privacy, but edge hardware is tightly constrained. Quantization makes models small enough to fit, at some cost to output quality. The paper asks what that trade actually costs: how quantization level affects **energy consumption, accuracy and latency** together, rather than one at a time.

APPROACH AND ARCHITECTURE

The authors benchmark 28 quantized models from the Ollama library on a Raspberry Pi 4 (4 GB). Prompts from five public datasets are sent through a local Ollama server; responses are scored by per-dataset evaluation scripts, while energy is captured by an external hardware power meter. Their MELODI tool wraps inference with energy monitoring.

The **cloud-computing dimension** is the deployment question itself: which inference work can be pushed to the edge rather than served centrally, and at what cost in energy and quality. The study is **text-only**; no multimodal component is involved.

02 Datasets

DATASET	SAMPLES USED	TYPE	USED HERE
CommonsenseQA	200	5-way multiple choice, commonsense	✅ yes
TruthfulQA (MC)	200	multiple choice, truthfulness	✅ yes
BIG-Bench-Hard	200	mixed reasoning	available, not yet run
GSM8K	200	grade-school maths, free text	available, not yet run
HumanEval	164	code generation	available, not yet run

All five are shipped as JSONL in the authors' data repository, pre-sampled from the NeurIPS 2024 Edge LLM Challenge selection. No train/validation split applies: every prompt is an evaluation item. Preprocessing is handled by the authors' `process_prompt()`, which wraps each question in a fixed instruction requiring the answer inside `<ANS></ANS>` tags. We changed no prompt text.

03 Environment

COMPONENT	PAPER	THIS REPRODUCTION
Device	Raspberry Pi 4, 4 GB RAM (CPU inference)	RTX 3050 Laptop GPU, 4 GB VRAM
OS	Linux	Windows 11 Home, build 26200
Driver	—	NVIDIA 581.83
Runtime	Ollama	Ollama 0.24.0
Python	3.x	3.13.5
Key libraries	not documented	ollama 0.6.3, jsonlines 4.0.0, pandas 3.0.6, scikit-learn 1.9.1, rouge_score 0.1.2, python-Levenshtein 0.27.5, nltk 3.10.3
Energy measurement	external hardware power meter	NVML software sampling (not comparable)
Protocol	`--repeat 5`, temperature 0	identical

04 Results: original vs reproduced

Accuracy is hardware-independent, so these figures are directly comparable. Each cell is 200 prompts × 5 repeats = **1,000 generations**, scored by the authors' own evaluation scripts. Paper values are from Table 8, p. 17.

MODEL	DATASET	PAPER	REPRODUCED	DIFF	P	VERDICT
qwen2.5-0.5b-q4_K_M	CommonsenseQA	0.32	0.291	−0.029	0.420	consistent
qwen2.5-1.5b-q4_K_M	CommonsenseQA	0.62	0.593	−0.027	0.474	consistent
qwen2.5-0.5b-q4_K_M	TruthfulQA	0.26	0.275	+0.015	0.660	consistent
qwen2.5-1.5b-q4_K_M	TruthfulQA	0.24	0.249	+0.009	0.786	consistent

> **All four reproducible cells agree with the published values.** Differences range from −0.029 to +0.015, and a two-proportion z-test returns p = 0.42–0.79 in every case: no statistically significant difference.

WHY SMALL DIFFERENCES REMAIN

- **Sampling.** Even at temperature 0, llama.cpp output is not bit-identical across builds and hardware; tie-breaking in the decoder differs.
- **Ollama version drift.** The paper does not pin an Ollama or model-blob version. Model tags are mutable, so the weights behind `qwen2.5:1.5b-instruct-q4_K_M` may have been re-uploaded since.
- **CPU vs GPU kernels.** Different quantized matrix-multiply kernels produce slightly different rounding.
- Differences are small, unsigned (two negative, two positive) and statistically insignificant — consistent with noise rather than a systematic error.

MEASUREMENTS BEYOND THE PAPER

Table 8 contains no q8_0 or fp16 rows for the 1.5B model. The authors' `exp.sh` explains why: on a 4 GB Raspberry Pi, 1.5B at q8_0 (1.6 GB) and fp16 (3.1 GB) were not run. These cells are therefore **new data, not reproductions**.

MODEL	COMMONSENSEQA	TRUTHFULQA
qwen2.5-1.5b-q4_K_M	0.593	0.249
qwen2.5-1.5b-q8_0	0.698	0.239
qwen2.5-1.5b-fp16	0.705	0.245

On CommonsenseQA, accuracy rises sharply from q4 to q8 (**+10.5 points**) and then plateaus at fp16 (+0.7). The paper could not observe this for the 1.5B model because its hardware could not hold the weights. On TruthfulQA the three are flat, so the effect is task-dependent.

05 Problems encountered and deviations

Six issues had to be resolved before the authors' code would run. All are recorded because several affect anyone attempting this reproduction.

#	PROBLEM	RESOLUTION
1	Data repository ships no requirements file. `scikit-learn`, `rouge_score`, `python-Levenshtein` and `nltk` are undocumented dependencies, each surfacing only when the previous one was installed.	Installed and documented in §03
2	`inference.py` uses `signal.SIGALRM` / `signal.alarm` for per-prompt timeouts. These are Unix-only and raise `AttributeError` on Windows.	Guarded with `hasattr(signal,"SIGALRM")`; Linux behaviour unchanged
3	`inference.py` prints every raw model response to stdout. Under the Windows cp1252 default encoding, a non-Latin-1 character raises `UnicodeEncodeError` and terminates the run. The 0.5B model emits CJK text often enough that its run died at 501/1000.	Forced `PYTHONIOENCODING=utf-8` in the child process rather than editing the script
4	**Silent data contamination.** Output is opened with `jsonlines.open(..., mode='a')`, so re-running a configuration *appends* rather than overwrites. After the crash in #3, the retry produced a 1,501-line file mixing a partial and a complete run, and the evaluator scored it without complaint.	Caught by asserting the generation count; file deleted and regenerated cleanly
5	Energy is measured in the paper with an external hardware power meter attached to a Raspberry Pi. We have no equivalent instrument and a different device class.	Energy and latency are declared **not comparable** and are not claimed
6	Model tags are unpinned, so the exact weights used by the authors cannot be guaranteed.	Recorded as a source of residual difference

Issues 2 and 3 indicate the artifacts have never been executed on Windows. Issue 4 is the most serious: it produces results that look valid and are silently wrong.

06 Candidate baseline evaluated and rejected

Before selecting this paper we attempted a reproduction of **Bench360: Benchmarking Local LLM Inference from 360 Degrees** (arXiv:2511.16682), and rejected it after getting it running.

- **Hardware.** Its experiment configs assume a 24 GB VRAM budget with 7B–32B models. Its four backends (vLLM, TGI, SGLang, LMDeploy) all preallocate far more memory than a 4 GB card has, so **no published cell could be reproduced** on available hardware.
- **Venue.** It is an unrefereed preprint, where the selection guidance prefers a strong venue.
- **Portability.** Its launcher is POSIX-only; Windows cannot spawn it. Model names containing `:` caused result CSVs to be written into NTFS alternate data streams, leaving 0-byte files while the run reported success.
- **Finding retained.** Despite targeting "consumer GPUs", none of its backends run on a real 4 GB consumer GPU. We added a GGUF/Ollama backend and obtained a working run, which is evidence of the same gap this project addresses.

Evaluating and discarding this candidate is itself part of the required reproducibility verification, and is reported rather than discarded.

07 Research gap

The reproduction establishes that the authors' accuracy results hold. Two limitations become visible from inside their own experimental design:

- **No sustained-load dimension.** Every measurement is a short run. Nothing in the study addresses what happens to a device held under continuous inference.

- **A hardware ceiling on their conclusions.** Their device could not hold the 1.5B model above q4, so their quantization guidance for that size is incomplete — as our new q8_0 and fp16 cells show, the missing region contains a 10.5-point accuracy gain.

> Their guidance implicitly assumes device throughput is stable. Our own measurements on the same GPU show it is not: under sustained load throughput falls **43%**, and the throughput advantage of q4 over q8 disappears entirely as the two converge to a common thermal floor.

So the quantization trade-off they characterise on a short-run CPU edge device **does not transfer to sustained GPU deployment**: the accuracy cost of aggressive quantization persists, while its speed benefit does not.

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08 Proposed method and implementation

Problem statement. The baseline characterises a configuration from a short measurement on a cool device. On a thermally constrained consumer GPU the device throttles under continuous load, so that figure overstates what the configuration sustains, and the size of the error depends on the quantization level. Deployment decisions taken from those numbers can be wrong.

Module. Rather than reporting the mean of a short window, fit the observed decay and extrapolate to the steady state:

```
tok/s(t) = a + b · exp(−t / tau)      predicted steady state = a
```

For a fixed `tau` the model is linear in `(a, b)`, so the implementation grid-searches `tau` over 20–600 s and solves the two-parameter least squares in closed form. It adds no dependencies and runs in milliseconds.

Two components are separable, which is what the ablation exploits:

- **A — sustained measurement.** Keep the device under continuous load for longer (300 s) instead of a short burst, and report the final window. Costs wall-clock time.
- **B — decay extrapolation.** Fit the curve above and report its asymptote instead of a window mean. Costs nothing.

Implementation: `src/thermal_estimator.py`. The original baseline is preserved: the “baseline” arm is exactly the short-window mean that the paper’s protocol produces, so every arm is computed from the same raw telemetry and the comparison is like-for-like.

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09 Evaluation and ablation

Ground truth is the mean throughput after 700 s of continuous load. Error is absolute percentage error against that value, averaged over the three GPU-resident configurations; fp16 is excluded from the averages because it is CPU-offloaded and barely throttles, which would flatter every method.

ARM	WHAT IT DOES	MAPE	MAX ERROR	COST
baseline	mean of first 60 s (the paper's regime)	51.1%	75.0%	60 s
+A	measure 300 s, report last 60 s	18.1%	27.7%	300 s
+B	extrapolate from first 60 s	17.9%	39.2%	60 s
+A+B	extrapolate from first 300 s	10.5%	23.3%	300 s

> Both components contribute, and they are not redundant. **B alone matches A's accuracy at one-fifth of the measurement cost** (17.9% vs 18.1%, 60 s vs 300 s), and combining them halves the error again to 10.5%. The ablation therefore attributes the improvement rather than assuming it.

PER-CONFIGURATION PREDICTIONS (TOK/S)

CONFIGURATION	ACTUAL	BASELINE	+A	+B	+A+B
qwen2.5-0.5b-q4_K_M	179.9	243.6	188.8	175.4	179.9
qwen2.5-1.5b-q4_K_M	66.2	115.8	84.5	58.3	81.6
qwen2.5-1.5b-q8_0	66.8	95.4	81.2	93.0	61.3
qwen2.5-1.5b-fp16 (CPU-offloaded)	24.6	25.3	22.1	24.7	17.8

The baseline overstates every GPU-resident configuration, by 43–75%. The failure is not uniform: it is largest for q4, the configuration a short benchmark would recommend. The fp16 row shows the mechanism from the other side — because a quarter of its work runs on the CPU it never reaches the GPU's thermal limit, so even the naive baseline is accurate for it.

One honest weakness: `+B` is unstable on q8 (93.0 against an actual 66.8, a 39% error) because 60 s of that run contains too little curvature to fit. `+A+B` corrects it. The extrapolation is therefore sensitive to how much decay is visible in the window, which is a limitation of the method rather than a tuning detail.

10 Final comparison

MEASURE	ORIGINAL PAPER	REPRODUCED BASELINE	PROPOSED METHOD
CommonsenseQA accuracy (1.5b-q4)	0.62	0.593	0.593
TruthfulQA accuracy (1.5b-q4)	0.24	0.249	0.249
Sustained-throughput error (MAPE)	not measured	51.1%	10.5%
Worst-case error	not measured	75.0%	23.3%
Measurement cost per configuration	short run	60 s	60 s (+B) / 300 s (+A+B)
Sustained-load characterisation	absent	absent	present
1.5B above q4 measured	no (4 GB Pi)	yes	yes

Accuracy is deliberately unchanged. The proposed module does not alter what the models answer, so the accuracy rows are identical to the reproduction; claiming an accuracy improvement would be false. The contribution is to the *performance characterisation*: the quantity the baseline reports is wrong by half under sustained load, and the module reduces that error roughly fivefold.

QUESTIONS THE COMPARISON ANSWERS

- **Reproduced successfully:** all four comparable accuracy cells, within noise (p = 0.42–0.79).
- **Not reproduced:** energy and latency — different device class and measurement instrument; three of five datasets and 24 of 28 models were out of scope.
- **Why differences arose:** non-deterministic decoding, unpinned model tags, CPU vs GPU quantized kernels. Small, unsigned, statistically insignificant.
- **Gap identified:** no sustained-load dimension, and a hardware ceiling that hid the 1.5B q8/fp16 region.
- **How the improvement addresses it:** it estimates the steady state the baseline never measures, from the same or a modestly longer measurement.
- **Evidence it works:** MAPE falls 51.1% → 10.5%, with an ablation separating the two components.
- **Remaining limitations:** see §11.

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11 Limitations

- Two of five datasets were run; BIG-Bench-Hard, GSM8K and HumanEval remain.
- Four of 28 models were run, chosen because they appear in the authors' `exp.sh` and fit in 4 GB.
- Energy and latency are **not** compared to the paper: different device class and measurement instrument.
- fp16 at 1.5B does not fit in 4 GB and ran at a 23%/77% CPU/GPU split, so its throughput belongs to a different regime, though its accuracy remains valid.
- Ambient temperature was not controlled during the sustained-load measurements.
- The estimator is evaluated on three GPU-resident configurations from single sustained runs. The sample is small, and the reported MAPE should be read as indicative rather than precise.
- Extrapolation quality depends on how much decay the measurement window contains; `+B` degrades when the window is nearly flat.
- A single device was used. Generalisation to other GPUs and thermal designs is untested.

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12 Reproducibility package

All code, configuration and raw outputs are public:

```
https://github.com/MFaheemS/Cloud-Project
```

- `reproduction/run_repro.py` — protocol driver (models × datasets, `--repeat 5`)
- `reproduction/finalize.py` — completeness check, evaluation, comparison table
- `reproduction/results/comparison_table.csv` — the table in §04
- `reproduction/patches/` — the two source changes made to the authors' code
- `src/`, `experiments/raw/` — thermal harness and 20 Hz telemetry underlying §07